Operating System

24k-0705 | Muhammad Adil Saeed | BCS-4j

Lab Task #01

```
student      7938  0.0  0.0  13024  3360 pts/0    R+   14:24   0:00 ps aux
student@Lab3W:~/Desktop$ touch file2.txt
student@Lab3W:~/Desktop$ ps aux > file2.txt
student@Lab3W:~/Desktop$ cat file2.txt
USER         PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root           1  0.0  0.0 167992 12156 ?        Ss   13:28   0:02 /sbin/init splash
```

```
student@Lab3W:~/Desktop$ mkdir 24k-0705
student@Lab3W:~/Desktop$ cd 24k-0705
student@Lab3W:~/Desktop/24k-0705$ date
Tue Jan 20 02:37:37 PM PKT 2026
student@Lab3W:~/Desktop/24k-0705$ uname -r
6.8.0-60-generic
student@Lab3W:~/Desktop/24k-0705$ arch
x86_64
student@Lab3W:~/Desktop/24k-0705$ lscpu
Architecture:            x86_64
  CPU op-mode(s):        32-bit, 64-bit
  Address sizes:         46 bits physical, 48 bits virtual
  Byte Order:            Little Endian
CPU(s):                  20
  On-line CPU(s) list:   0-19
Vendor ID:               GenuineIntel
  Model name:            12th Gen Intel(R) Core(TM) i7-12700
    CPU family:          6
    Model:               151
    Thread(s) per core:  2
    Core(s) per socket:  12
    Socket(s):           1
    Stepping:            2
    CPU max MHz:         4900.0000
    CPU min MHz:         800.0000
    BogoMIPS:            4224.00
    Flags:               fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush dts acpi mmx fxsr sse sse2 ss ht tm pbe syscall nx pdpe1gb rdtscp lm constant_tsc art arch_pe
                          pebs bts rep_good nopl xtopology nonstop_tsc cpuid aperfmperf tsc_known_freq pni pclmulqdq dtes64 monitor ds_cpl vmx smx est tm2 ssse3 sdbg fma cx16 xtpr pdcm sse4_1 sse4_2
                         ic movbe popcnt tsc_deadline_timer aes xsave avx f16c rdrand lahf_lm abm 3dnowprefetch cpuid_fault epb ssbd ibrs ibpb stibp ibrs_enhanced tpr_shadow flexpriority ept vpid ep
                         fsgsbase tsc_adjust bmi1 avx2 smep bmi2 erms invpcid rdseed adx smap clflushopt clwb intel_pt sha_ni xsaveopt xsavec xgetbv1 xsaves split_lock_detect user_shstk avx_vnni dth
                         da arat pln pts hwp hwp_notify hwp_act_window hwp_epp hwp_pkg_req hfi vnmi umip pku ospke waitpkg gfni vaes vpclmulqdq tme rdpid movdiri movdir64b fsrm md_clear serialize pc
                          arch_lbr ibt flush_l1d arch_capabilities
Virtualization features:
  Virtualization:        VT-x
Caches (sum of all):
  L1d:                   512 KiB (12 instances)
  L1i:                   512 KiB (12 instances)
  L2:                    12 MiB (9 instances)
  L3:                    25 MiB (1 instance)
NUMA:
  NUMA node(s):          1
  NUMA node0 CPU(s):     0-19
Vulnerabilities:
  Gather data sampling:  Not affected
  Itlb multihit:         Not affected
  L1tf:                  Not affected
  Mds:                   Not affected
  Meltdown:              Not affected
  Mmio stale data:       Not affected
  Reg file data sampling: Mitigation; Clear Register File
  Retbleed:              Not affected
  Spec rstack overflow:  Not affected
  Spec store bypass:     Mitigation; Speculative Store Bypass disabled via prctl
  Spectre v1:            Mitigation; usercopy/swapgs barriers and __user pointer sanitization
```

```
                         ic movbe popcnt tsc_deadline_timer aes xsave avx f16c rdrand lahf_lm abm 3dnowprefetch cpuid_fault epb ssbd ibrs ibpb stibp ibrs_enhanced tpr_shadow flexpriority ept vpid ep
                         fsgsbase tsc_adjust bmi1 avx2 smep bmi2 erms invpcid rdseed adx smap clflushopt clwb intel_pt sha_ni xsaveopt xsavec xgetbv1 xsaves split_lock_detect user_shstk avx_vnni dth
                         da arat pln pts hwp hwp_notify hwp_act_window hwp_epp hwp_pkg_req hfi vnmi umip pku ospke waitpkg gfni vaes vpclmulqdq tme rdpid movdiri movdir64b fsrm md_clear serialize pc
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  Spec rstack overflow:  Not affected
  Spec store bypass:     Mitigation; Speculative Store Bypass disabled via prctl
  Spectre v1:            Mitigation; usercopy/swapgs barriers and __user pointer sanitization
  Spectre v2:            Mitigation; Enhanced / Automatic IBRS; IBPB conditional; RSB filling; PBRSB-eIBRS SW sequence; BHI BHI_DIS_S
  Srbds:                 Not affected
  Tsx async abort:       Not affected
student@Lab3W:~/Desktop/24k-0705$ date
Tue Jan 20 02:38:20 PM PKT 2026
student@Lab3W:~/Desktop/24k-0705$ sudo date 12121200
[sudo] password for student:
Sat Dec 12 12:00:00 PM PKT 2026
student@Lab3W:~/Desktop/24k-0705$ ls -l
total 0
student@Lab3W:~/Desktop/24k-0705$ ls -a

student@Lab3W:~/Desktop/24k-0705$ ls grep [1]
ls: cannot access 'grep': No such file or directory
ls: cannot access '[1]': No such file or directory
student@Lab3W:~/Desktop/24k-0705$ mkdir dir1
student@Lab3W:~/Desktop/24k-0705$ mkdir dir2 dir3
student@Lab3W:~/Desktop/24k-0705$ pwd
/home/student/Desktop/24k-0705
student@Lab3W:~/Desktop/24k-0705$ rm -r dir1
student@Lab3W:~/Desktop/24k-0705$ rmdir dir2
student@Lab3W:~/Desktop/24k-0705$ touch -c -t 212320202323 file_or_dir
touch: invalid date format '212320202323'
student@Lab3W:~/Desktop/24k-0705$ touch -c -t 202601201200 file_or_dir
student@Lab3W:~/Desktop/24k-0705$ pwd
/home/student/Desktop/24k-0705
student@Lab3W:~/Desktop/24k-0705$ ls
dir3
```